

# Safety, tolerance, and metabolism of broccoli sprout glucosinolates and isothiocyanates: a clinical phase I study.

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Broccoli sprouts are widely consumed in many parts of the world. There have been no reported concerns with respect to their tolerance and safety in humans. A formal phase I study of safety, tolerance, and pharmacokinetics appeared justified because these sprouts are being used as vehicles for the delivery of the glucosinolate glucoraphanin and its cognate isothiocyanate sulforaphane [1-isothiocyanato-(4R)-(methylsulfinyl)butane] in clinical trials. Such trials have been designed to evaluate protective efficacy against development of neoplastic and other diseases. A placebo-controlled, double-blind, randomized clinical study of sprout extracts containing either glucosinolates (principally glucoraphanin, the precursor of sulforaphane) or isothiocyanates (principally sulforaphane) was conducted on healthy volunteers who were in-patients on our clinical research unit. The subjects were studied in three cohorts, each comprising three treated individuals and one placebo recipient. Following a 5-day acclimatization period on a crucifer-free diet, the broccoli sprout extracts were administered orally at 8-h intervals for 7 days (21 doses), and the subjects were monitored during this period and for 3 days after the last treatment. Doses were 25 micromol of glucosinolate (cohort A), 100 micromol of glucosinolate (cohort B), or 25 micromol of isothiocyanate (cohort C). The mean cumulative excretion of dithiocarbamates as a fraction of dose was very similar in cohorts A and B (17.8 +/- 8.6% and 19.6 +/- 11.7% of dose, respectively) and very much higher and more consistent in cohort C (70.6 +/- 2.0% of dose). Thirty-two types of hematology or chemistry tests were done before, during, and after the treatment period. Indicators of liver (transaminases) and thyroid [thyroid-stimulating hormone, total triiodothyronine (T3), and free thyroxine (T4)] function were examined in detail. No significant or consistent subjective or objective abnormal events (toxicities) associated with any of the sprout extract ingestions were observed.